

# SINGYU MUK

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## RESEARCH INTERESTS

Embodied AI and scalable robot learning from human/unlabelled video and synthetic 3D data; transferable motion and interaction representations; motion generation for embodied agents; multimodal and video generative models.

## EDUCATION

### Master of Artificial Intelligence, Monash University

Nov 2025

Melbourne, Australia | First Class Honours | WAM 81/100 | Research Pathway

**Master's Research Project:** *Australian Sign Language Generation*; developed multi-view 3D motion reconstruction and MM-WLAuslan-MV3D, later incorporated into MV-Sign (WACV 2027).

### Bachelor of Arts in Leadership, Trinity Western University

Dec 2017

Vancouver, Canada

## PUBLICATIONS

[1] Singyu Muk, Kaustubh Kundu, Kunyuan Xie, Weijie Wang, Kalin Stefanov. *MV-Sign: Multi-View 3D Motion Integration and Enhanced Text-to-Motion Conditioning for Sign Language Production*. **Accepted, WACV 2027**. [OpenReview](#)

[2] Singyu Muk, Xuanren Song, Kalin Stefanov. *S2HAND: Accuracy-Preservation Trade-Offs for Hand Specialization in Whole-Body Pose Models*. **Under review, WACV 2027**. [OpenReview](#)

## RESEARCH EXPERIENCE

### AI Research Engineer / Research Student | Monash Assistive Tech, Monash University

Mar 2025 - Present

Melbourne, Australia | Supervisor: Kalin Stefanov

- Ongoing research extends motion generation toward embodied agents, focusing on facial-expression motion generation and the use of video-generation models as motion priors or conditioning signals for controllable embodied behavior.
- First author of MV-Sign (accepted, WACV 2027): developed a multi-view SMPL-X curation pipeline using AiOS/WiLoR, spatial-temporal alignment, non-sign filtering, and cross-view body/finger repair; released How2Sign-MV3D, ASLCitizen-3D, MM-WLAuslan-MV3D, and SGNify-MV.
- Designed a sign-aware spatio-temporal VQ-VAE with joint body-hand encoding, pseudo-gloss conditioning, and keyframe-aware retrieval, reducing body and hand DTW-PA-JPE by 32.8% and 20.9% versus the reported state of the art on How2Sign.
- First author of S2HAND (under review, WACV 2027): specialized Sapiens2-5B for hand pose using protected-output adaptation and Fisher-selected Transformer blocks; raised three-set macro AUC from 57.83% to 65.73% while preserving whole-body and segmentation behavior.

### AI Internship - Video Generation & Motion Data Pipeline | Chinese Academy of Sciences

Nov 2024 - Feb 2025

Digital Content Synthesis and Forgery Detection Laboratory | Supervisor: Fan Tang

- Built a video-to-pose extraction and motion-data preparation pipeline processing ~90K videos; implemented pose denoising and correction that reduced a jerk-based motion shakiness score by ~20%.

## TECHNICAL SKILLS

**Machine Learning:** Python, PyTorch, CUDA, VQ-VAE, Transformers, video generation models. **Robotics / Embodied AI:** CALVIN benchmark; robot-learning dataset and benchmark workflows. **3D / Vision:** SMPL-X, AiOS, WiLoR, Blender, 3D pose/motion reconstruction, synthetic-data pipelines. **Systems / Research Tools:** Linux, Slurm, Git, LaTeX, Docker, AWS.

## ACHIEVEMENTS

- Ranked Top 5 in a Monash University deep-learning image-recognition Kaggle competition (cohort ranking).